

Deep Learning and Causal AI for Estimation

Lecture 1: Feedforward Neural Networks

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Introduction

Neural network models have driven many of the important breakthroughs in machine learning and AI over the last ten years.

They were already well-known and popular in the 1990s.

Several interrelated drivers of their recent growth:

1. Availability of large datasets for training
2. Hardware advances: GPU/TPU, cloud infrastructure
3. Software advances: high-level frameworks for model estimation that hide many details from the user
4. Algorithmic advances: better training algorithms

In this lecture, we describe the structure of the simplest class of neural networks: feed-forward neural networks.

Review of Supervised Learning

Begin with observations $(y_i, \mathbf{x}_i)_{i=1}^N$.

Function $f_\theta(\mathbf{x})$ (potentially non-linear) that maps inputs into outputs.

θ is a vector of parameters that determines the shape of the function.

Typical workflow is

1. Split data into training T and test E portions
2. Fit $f_\theta(\mathbf{x})$ on training data to minimize loss function

$$\theta^* = \arg \min_{\theta} \sum_{i \in T} L(f_\theta(\mathbf{x}_i), y_i)$$

3. Evaluate test-set loss

$$\sum_{i \in E} L(f_{\theta^*}(\mathbf{x}_i), y_i)$$

Outline

Basic Architecture of a Neural Network

Estimation

A neuron

As usual, suppose that each data observation i has J associated covariates which we stack in $\mathbf{x}_i$.

As in linear regression, we can linearly combine covariates and add a constant:

$$z_i = \theta_0 + \sum_{j=1}^J \theta_j x_{i,j}.$$

In neural network jargon, the $\{\theta_j\}_{j=1}^J$ terms are sometimes called *weights* and θ_0 is called a *bias*.

We can transform the linear combination via an *activation function* ϕ .

Activation Functions

Traditionally:

1. A sigmoid function (logistic regression):

$$\phi(z_i) = \frac{1}{1 + e^{-z_i}}$$

2. A particular limiting case: step function.
3. Hyperbolic tangent.
4. Identity function (linear regression): $\phi(z_i) = z_i$.

Other activation functions have gained popularity recently:

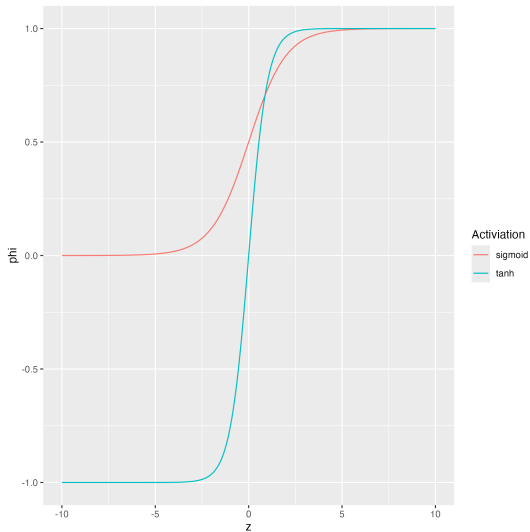
1. Rectified linear unit (ReLU):

$$\phi(z_i) = \max(0, z_i)$$

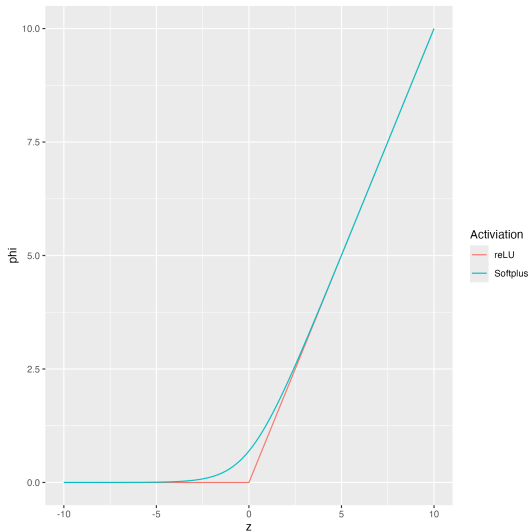
2. Softplus:

$$\phi(z_i) = \log(1 + e^{z_i})$$

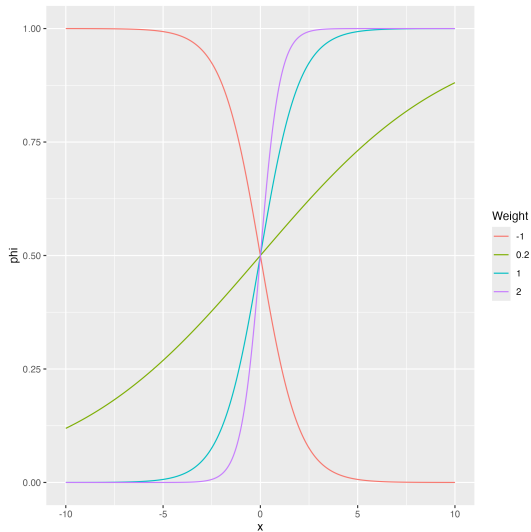
Activation Function Plots I



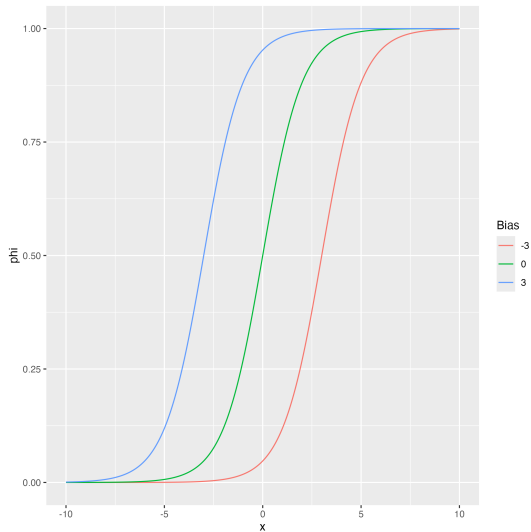
Activation Function Plots II



Effect of Varying Weight (Sigmoid)



Effect of Varying Bias (Sigmoid)



Combining Neurons into a Neural Network

Now suppose we have K separate activation functions operating over the same input $\mathbf{x}_i$.

We first obtain K separate linear combinations of covariates

$$z_{i,k} = \theta_{0,k} + \sum_{j=1}^J \theta_{j,k} x_{i,j} \quad \forall k = 1, \dots, K \quad (\text{Hidden Layer})$$

Next we linearly combine each separate neuron to create a final output

$$f_{\theta}(\mathbf{x}_i) = \theta_0 + \sum_k \theta_k \phi(z_{i,k}) \quad (\text{Output Layer})$$

This is known as a single layer network.

Consists of K non-linear transformations of linear combinations of input variables.

Matrix Notation

We can write the single-layer network compactly in matrix form.

Stack the weights into a $K \times J$ matrix $\mathbf{W}$ and the biases into $\mathbf{b} \in \mathbb{R}^K$:

$$\mathbf{h}_i = \phi(\mathbf{W}\mathbf{x}_i + \mathbf{b}) \quad (\text{Hidden Layer})$$

$$f(\mathbf{x}_i \mid \boldsymbol{\theta}) = \mathbf{w}^\top \mathbf{h}_i + b_0 \quad (\text{Output Layer})$$

where ϕ is applied element-wise and $\mathbf{w} \in \mathbb{R}^K$, $b_0 \in \mathbb{R}$ are the output layer parameters.

Classic Result I

Result

Universal approximation theorem: [Hornik et al., 1989] A neural network with at least one hidden layer can approximate any Borel measurable function mapping finite-dimensional spaces to any desired degree of accuracy.

Why Go Deep?

The universal approximation theorem guarantees that a single hidden layer *suffices* — but with important caveats:

1. The required width K may be **exponentially large** in the input dimension J .
2. Existence $\neq$ efficient learnability. The theorem says nothing about whether we can find the right weights in practice.

An alternative to making a network wider is to make it *deeper*: stack multiple hidden layers so that each layer builds on features learned by the previous one.

In practice, adding layers provides a much more efficient parameterization than adding neurons to a single layer.

Adding Layers

Additional layers can easily be added. Denote the output of layer l as $\mathbf{h}_i^{(l)}$, with $\mathbf{h}_i^{(0)} = \mathbf{x}_i$:

$$\mathbf{h}_i^{(1)} = \phi\left(\mathbf{W}^{(1)}\mathbf{x}_i + \mathbf{b}^{(1)}\right) \quad (\text{Hidden Layer 1})$$

$$\mathbf{h}_i^{(2)} = \phi\left(\mathbf{W}^{(2)}\mathbf{h}_i^{(1)} + \mathbf{b}^{(2)}\right) \quad (\text{Hidden Layer 2})$$

$$\vdots$$

$$f_{\theta}(\mathbf{x}_i) = \mathbf{w}^{(L+1)\top} \mathbf{h}_i^{(L)} + b^{(L+1)} \quad (\text{Output Layer})$$

where $\mathbf{W}^{(l)}$ is a $K_l \times K_{l-1}$ weight matrix, $\mathbf{b}^{(l)} \in \mathbb{R}^{K_l}$ is a bias vector, and ϕ is applied element-wise. For the first hidden layer, $K_0 = J$ since its input is $\mathbf{x}_i \in \mathbb{R}^J$.

Overall structure is called a *feed-forward neural network*.

Multiple layers allow for complex downstream structures to develop by combining simpler upstream structures.

Multi-Valued Output Layer

More generally, the output layer can produce M outputs:

$$\mathbf{f}_{\theta}(\mathbf{x}_i) = \mathbf{W}^{(L+1)}\mathbf{h}_i^{(L)} + \mathbf{b}^{(L+1)} \in \mathbb{R}^M$$

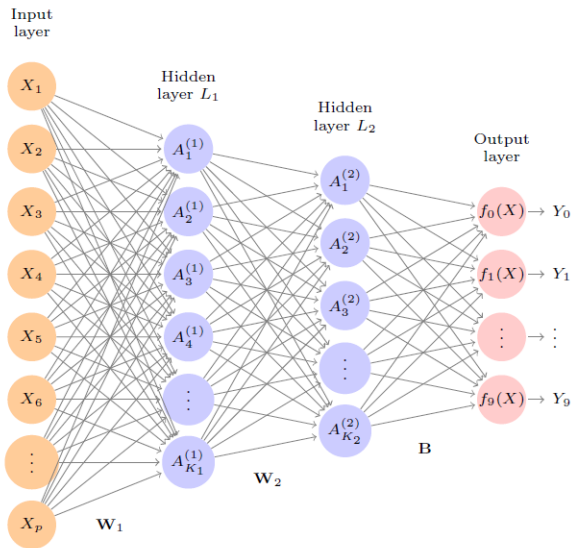
Two common uses:

1. Real-valued outputs: each f_m predicts a different target variable.
2. Class probabilities via the softmax function:

$$P_m(\mathbf{x}_i) = \frac{\exp(f_m)}{\sum_{m'=1}^M \exp(f_{m'})}$$

where M is the number of classes to predict.

Graphical Representation



Why Does Depth Help?

[Montúfar et al., 2014]: a ReLU network with L layers of K neurons each can produce $O(K^L)$ linear regions of the input space, versus $O(K \cdot L)$ for a single hidden layer with the same total neurons.

[Telgarsky, 2016]: there exist functions exactly representable by a depth- L network that cannot be approximated at depth $L - 1$ without exponentially more neurons.

Bottom line: depth is an exponentially more efficient use of parameters than width.

Outline

Basic Architecture of a Neural Network

Estimation

Loss Functions and Gradient Descent

Recall that we want to find $\theta^* = \arg \min_{\theta} \mathcal{L}(\theta)$ where $\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N L(f_{\theta}(\mathbf{x}_i), y_i)$.

Typical choices for L :

1. Regression (mean squared error): $L = (y_i - f_{\theta}(\mathbf{x}_i))^2$
2. Classification (cross-entropy): $L = - \sum_{m=1}^M y_{i,m} \log P_m(\mathbf{x}_i \mid \theta)$

Gradient descent: repeatedly step in the direction of steepest descent:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta)$$

where $\eta > 0$ is the *learning rate* (a small step size).

Stochastic Gradient Descent

Computing $\nabla_{\theta} \mathcal{L}(\theta)$ requires summing over all N data points, which is expensive for large datasets.

Mini-batch SGD: instead approximate the gradient using a random batch $\mathcal{B}$ of B observations:

$$\theta \leftarrow \theta - \frac{\eta}{B} \sum_{i \in \mathcal{B}} \nabla_{\theta} L(f_{\theta}(\mathbf{x}_i), y_i)$$

This is an unbiased estimate of the full gradient, but much cheaper to compute.

An *epoch* is a single pass through all of the training data.

Automatic Differentiation

SGD requires computing $\nabla_{\theta}\mathcal{L}$, the gradient of the loss with respect to all network parameters.

Since a neural network is a composition of differentiable operations, the chain rule gives an exact gradient. *Automatic differentiation* is the algorithm that evaluates this efficiently.

Modern frameworks (JAX, PyTorch) make this a single function call.

References I

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